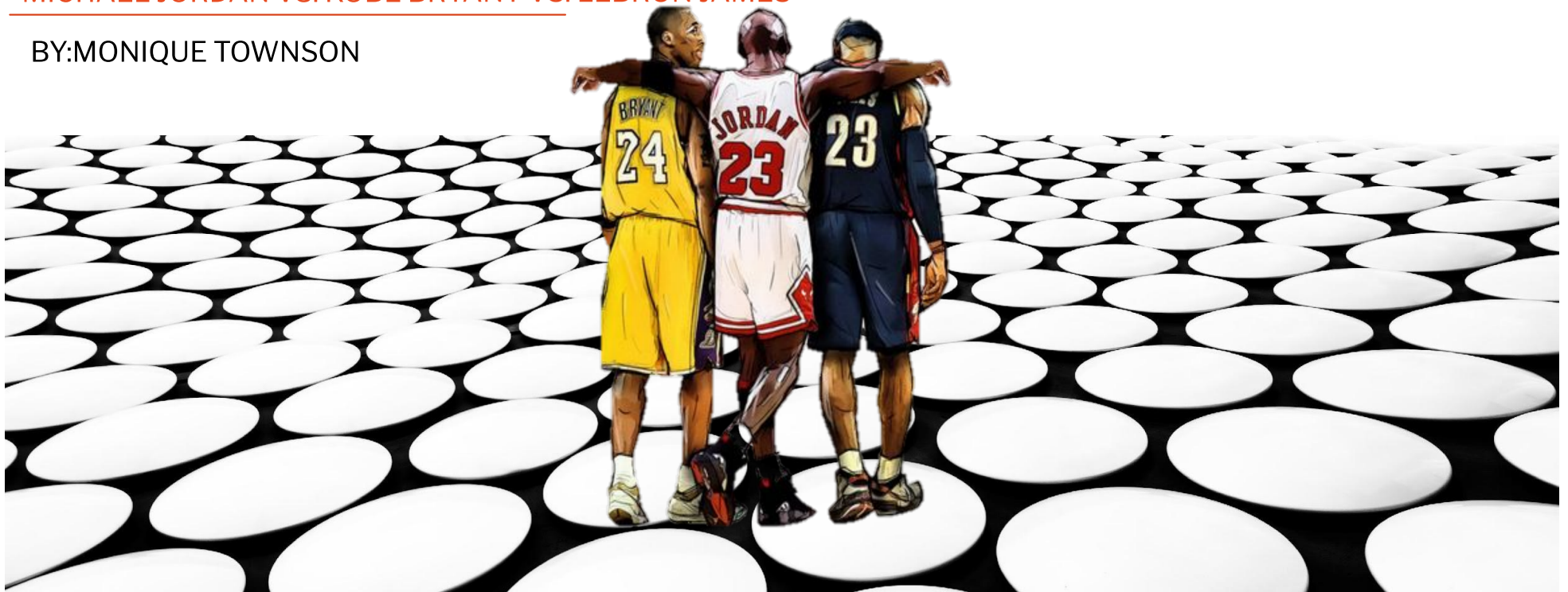

WHO'S THE GREATEST OF ALL TIME (GOAT)?

MICHAEL JORDAN VS. KOBE BRYANT VS. LEBRON JAMES

BY: MONIQUE TOWNSON



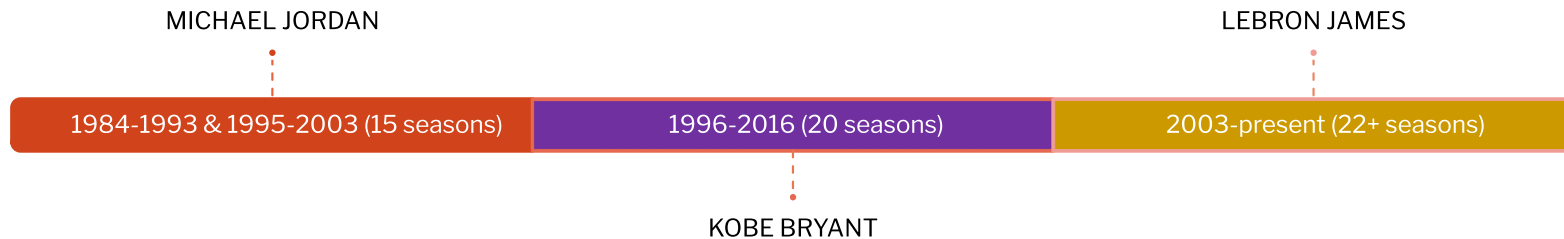
DATASETS

- All Games Stats Dataset: Found on the [Kaggle website](#)**

This dataset contains detailed per-game statistics spanning multiple seasons for each player. Key variables include field goals made and attempted, total points, assists, rebounds, steals, blocks, turnovers, and game outcomes (wins/losses). It provides a view of each player's game-by-game performance and scoring efficiency.

- Advanced Stats Dataset: Found on the [Kaggle website](#)**

This dataset offers advanced performance metrics that capture a player's overall impact and efficiency beyond traditional box score stats. Variables analyzed include true shooting percentage (TS%), usage rate, win shares, box plus-minus (BPM), and distribution patterns of points, rebounds, assists, steals, blocks, and turnovers



CLEANING DATA



IDENTIFY MISSING DATA &
INCOMPLETE ROWS



REMOVED ANY ROWS WITH
MISSING OR "0" GAMES
PLAYED



FILTERED OUT UNNECESSARY
COLUMNS AND DOUBLE CHECK
FORMATTING

EVALUATION OF INFORMATION

To determine who is the greatest basketball player of all time, it's essential to analyze a variety of key performance metrics that capture different aspects of their impact on the game. The selected variables provide a comprehensive view of each player's offensive efficiency, scoring ability, playmaking, defensive contributions, and overall effectiveness:

Player	Points/ Game (PTS)	Field Goal Avg. (FG.)	Three Point Avg. (X3P.)	Total Rebounds (TRB)	Assist (AST)	Steals (STL)	Turnovers (TOV)
Kobe Bryant	25.08	44.41%	30.01%	5.21	4.69	1.44	2.97
Lebron James	27.44	50.33%	30.95%	7.66	7.20	1.64	3.50
Michael Jordan	30.44	49.39%	27.89%	6.25	5.32	2.26	2.77

Kobe Bryant

Trails slightly in efficiency and playmaking but remains a strong scorer. Can possibly show a high impact in other areas.

LeBron James

Excels in shooting efficiency, rebounds, assists, and overall game impact, reflecting his all-around versatility

Michael Jordan

Leads in scoring and steals, highlighting his offensive dominance and defensive impact



HOW TO DETERMINE THE GREATEST PLAYER?

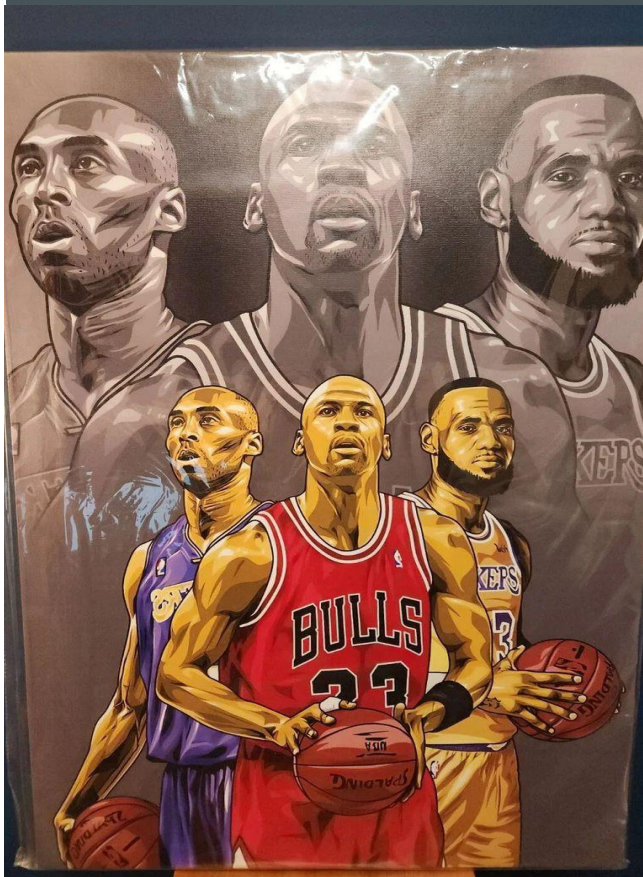
IDENTIFYING METRICS TO ANALYZE

- **Shooting Efficiency:** Examined field goals made vs. attempted and true shooting percentage to assess scoring efficiency.
- **Overall Contribution:** Compared total points, assists, rebounds, steals, blocks, and turnovers to understand each player's all-around impact.
- **Advanced Metrics:** Used win shares and box plus-minus to quantify each player's contribution to team success and overall value.
- **Usage and Distribution:** Analyzed usage rates and the distribution of key stats to understand how players influence the game and their roles within their teams.
- **Win/Loss Impact:** Investigated game outcomes to evaluate the players' impact on team success.



PLAYER IMPACT

- OFFENSIVE IMPACT
- DEFENSIVE IMPACT
- FIELD GOALS (FG)
- PLAYER EFFICIENCY RATING (PER)

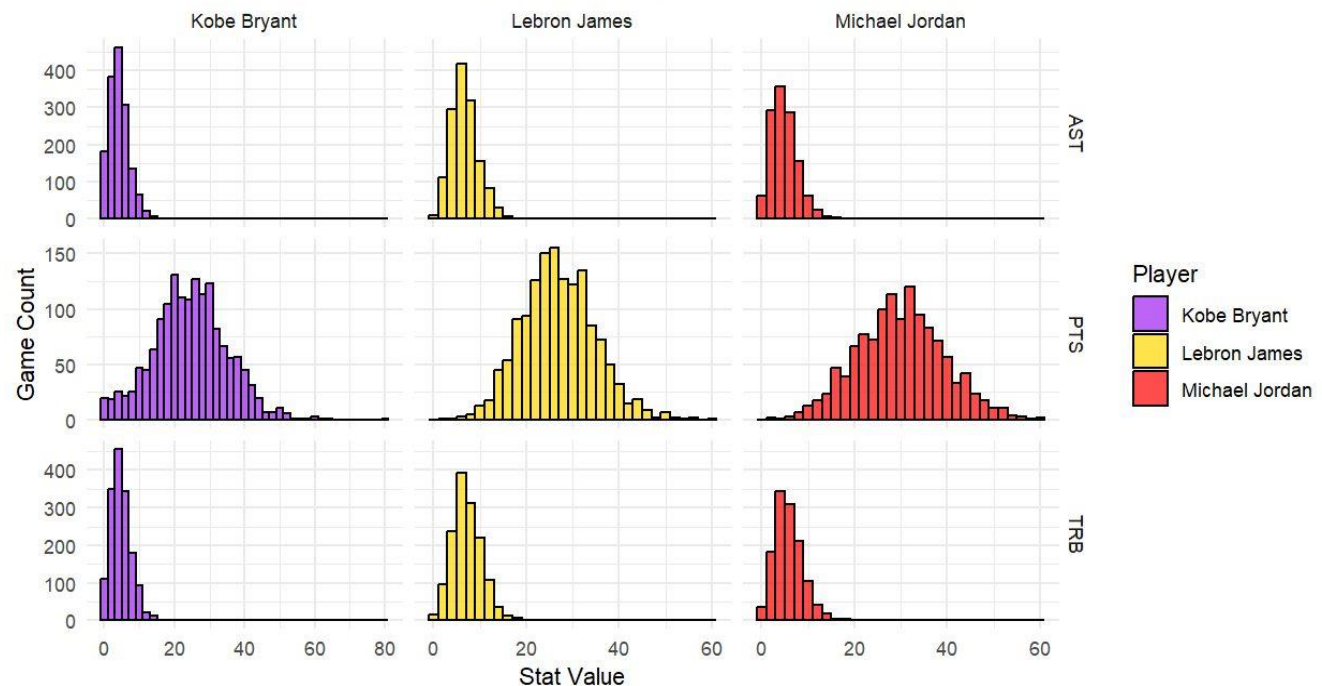


Kobe Bryant (Purple Bars)

Maintained strong but slightly lower averages in scoring, assists, and rebounds, highlighting his role as a primary scorer with a more focused contribution profile.

OFFENSIVE IMPACT

Distribution of Points, Rebounds, and Assists by Player

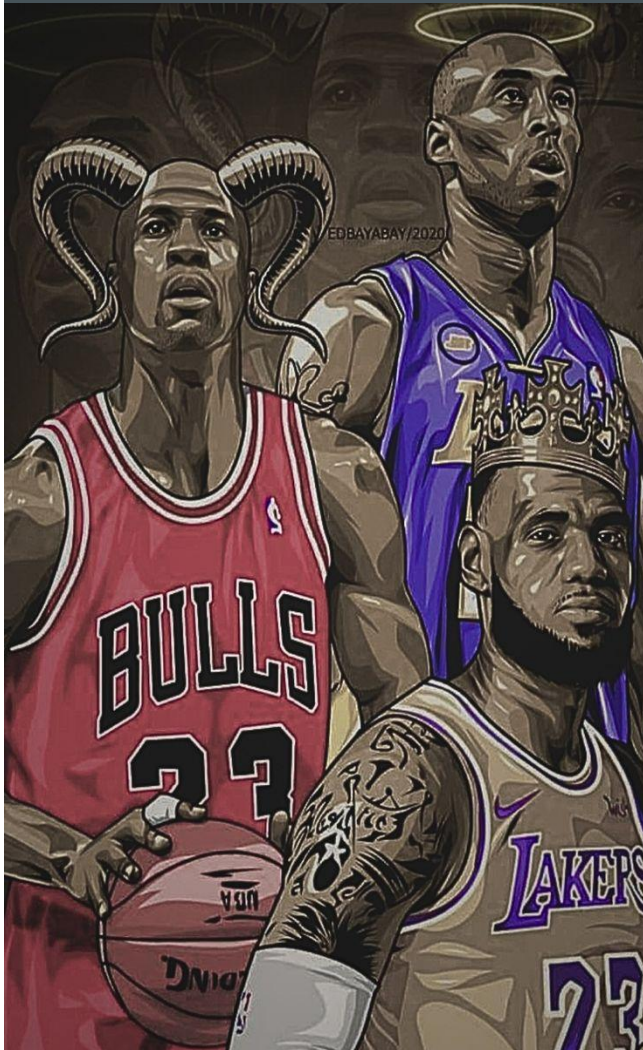


LeBron James (Yellow Bars)

Demonstrated the most well-rounded impact, leading in both assists and rebounds, reflecting his versatility and all-around contribution to team play.

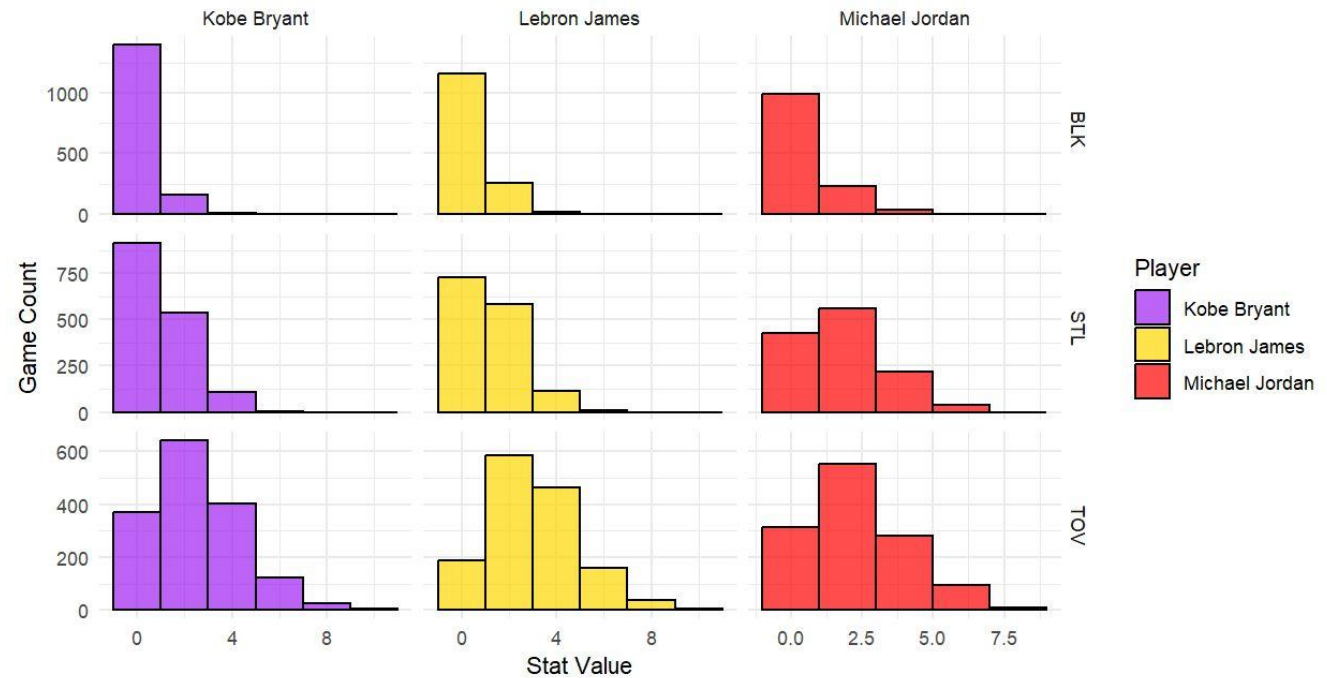
Michael Jordan (Red Bars)

Led in scoring with the highest median points per game, showcasing his dominant offensive role across a wide range of high-scoring performances.



DEFFENSIVE IMPACT

Distribution of Steals, Blocks, and Turnovers



Kobe Bryant (Purple Bars)

Contributed consistently on defense with moderate steals and blocks, while maintaining a controlled turnover rate compared to LeBron.

LeBron James (Yellow Bars)

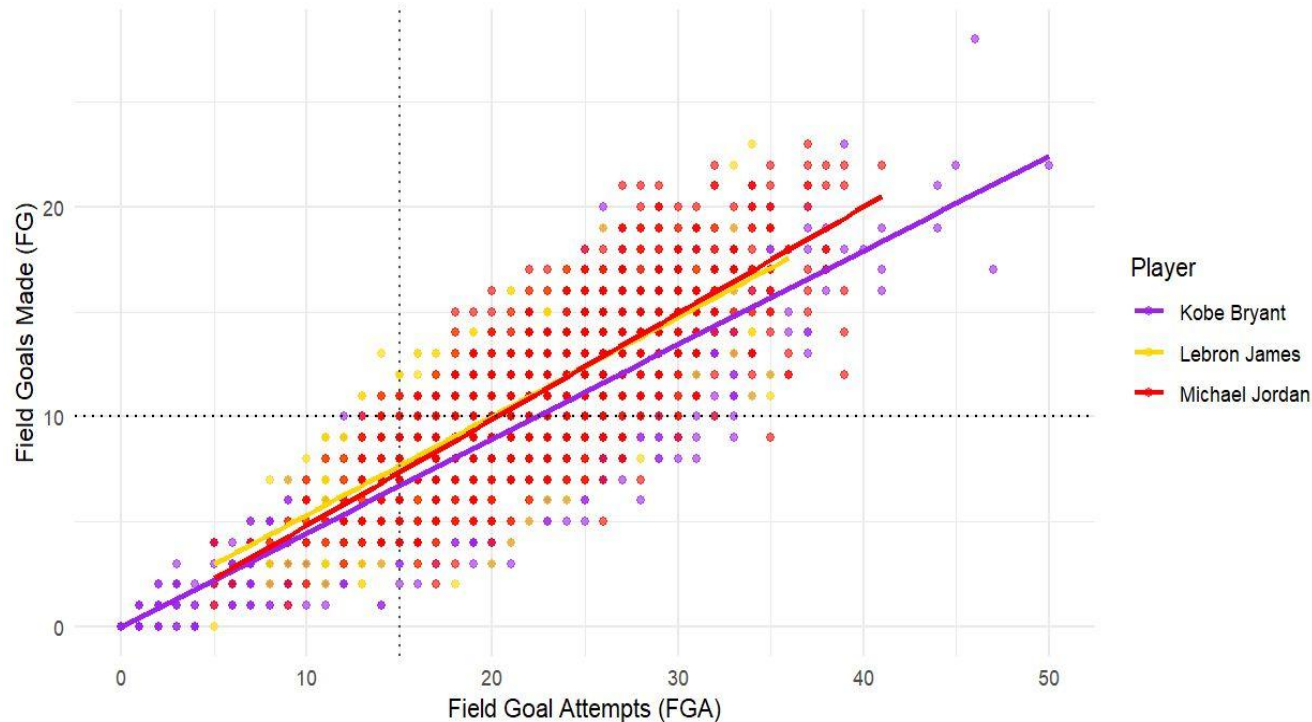
Showed balanced defensive stats with solid steal and block numbers but had the highest turnover rate due to his high playmaking role.

Michael Jordan (Red Bars)

Led in steals and had low turnovers, highlighting his defensive sharpness and ball control

FG vs. FGA with Regression Lines for 3 Players

Includes reference lines at FG = 10 and FGA = 15



Kobe Bryant (Purple Dots/Line)

Kobe's regression line has a positive slope but slightly less steep than Jordan's and LeBron's, suggesting solid but comparatively lower shooting efficiency, especially as attempts increase

LeBron James (Yellow Dots/Line)

LeBron's regression line shows the steepest slope among the three, indicating his high scoring efficiency even at higher volumes of shot attempts, emphasizing his ability to maintain efficiency as usage increases.



FIELD GOALS (FG)

The regression line on this scatter represents the relationship between how many shots each player attempts and how many they make. The **horizontal reference** line marks the level where a player scores 10 field goals in a game. The **vertical reference** line marks where a player attempts 15 field goals in a game.

Michael Jordan (Red Dots/Line)

Jordan's regression line has a strong positive slope, reflecting his consistent ability to convert a high proportion of shot attempts into made field goals, demonstrating efficient scoring regardless of volume.

PLAYER EFFICIENCY RATING (PER)

Is a single-number statistic that summarizes the player's overall per-minute performance. It accounts for both positive contributions (like points, assists, rebounds, steals, and blocks) and negative ones (like missed shots and turnovers), and is adjusted for game pace. A PER of **15** is the NBA league average. Higher values indicate greater efficiency and overall impact.



Kobe Bryant (Purple Box)

Kobe Bryant's PER is noticeably lower on average, with a median closer to 23–24 and a wider range extending to some low values, indicating more variability and generally less efficiency compared to the other two legends.

LeBron James (Yellow Box)

LeBron's PER has a slightly wider spread with some higher outliers; suggesting occasional peak performances above Jordan's range

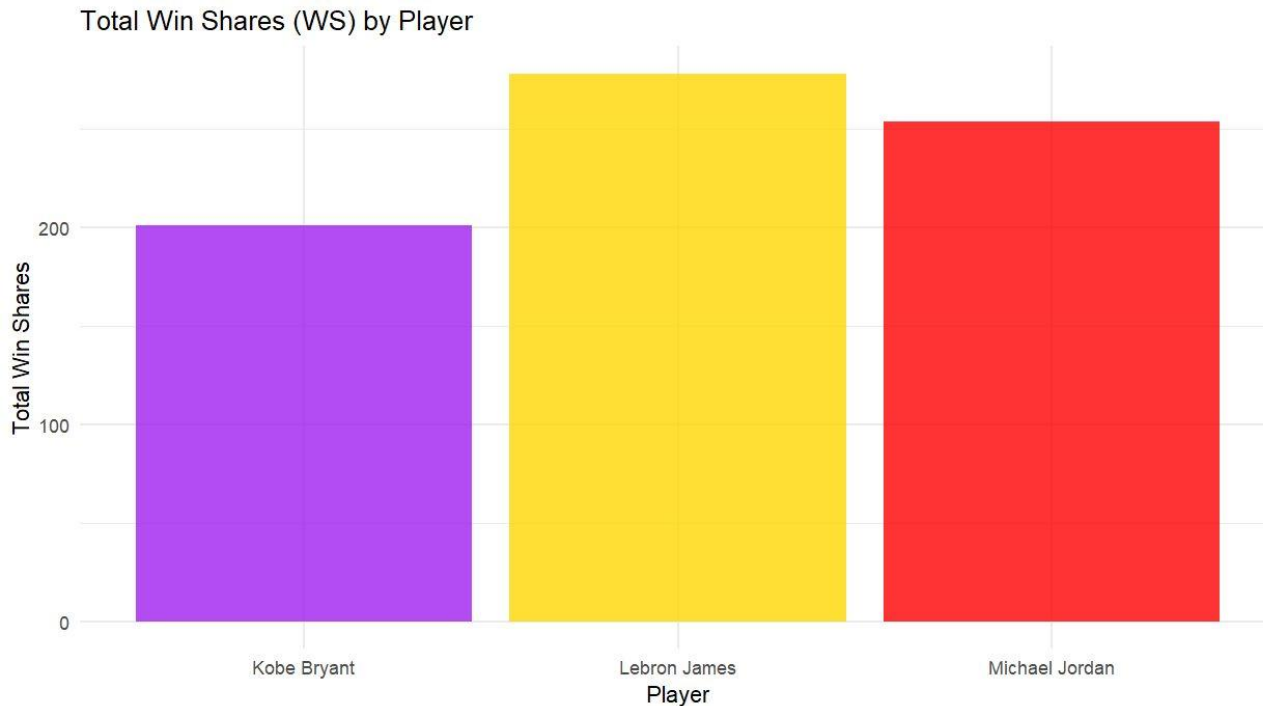
Michael Jordan (Red Box)

Michael Jordan and LeBron James have very similar median Player Efficiency Ratings (PER), both hovering around 28–29, indicating consistently high performance levels



TEAM IMPACT

- WIN SHARES (WS)
- USAGE RATE (USG)
- TRUE SHOOTING PERCENTAGE (TS)
- WINS & LOSSES



Kobe Bryant (Purple Bar)

While a dominant player, Kobe has a lower total WS, which can be attributed to both a shorter peak and the natural effects of injuries and team dynamics over his career.

LeBron James (Yellow Bar)

Higher total WS reflects both his sustained elite performance and his longer career span—playing more seasons and minutes compared to Jordan and Kobe.

Michael Jordan (Red Bar)

Despite playing fewer seasons (including a couple of retirements), still ranks second, demonstrating exceptional impact during his peak years.



WIN SHARES (WS)

Is an advanced basketball metric that estimates how many wins a player contributed to their team over a career. It breaks down the player's overall impact into **offensive** and **defensive contributions**, combining them into a single number representing total value.

USAGE RATE (USG)

Estimates the percentage of team plays used by a player while they are on the court. It accounts for: Field goal attempts, free-throw attempts and turnovers. Essentially, it measures how involved a player is in their team's offense. A higher USG% means the player is more central to the offense when on the floor.



Kobe Bryant (Purple Bars)

Distribution is broad, ranging from low 20s to upper 30s. Indicates variation over his career (early years, peak, post-prime), but with a notable concentration around 30–34%.

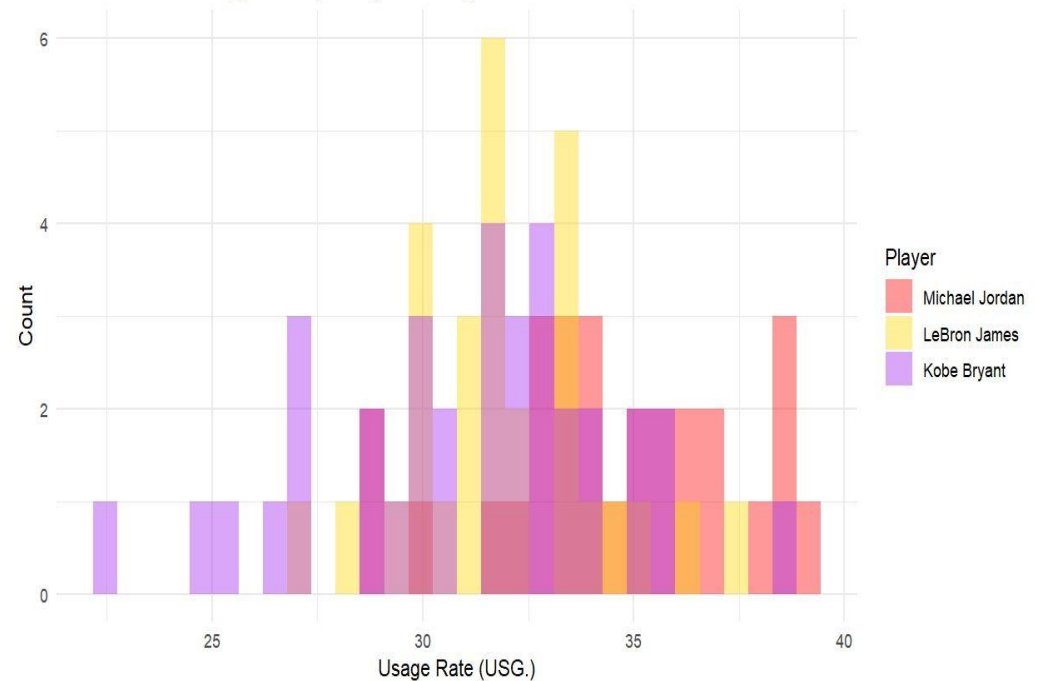
LeBron James (Yellow Bars)

More centered around 30–33%, with fewer outliers at the high end. Suggests he shared the ball more compared to Jordan, reflecting his passing-heavy style.

Michael Jordan (Red Bars)

USG% frequently ranges between 30–35%, with some seasons even near 38–39% (very high usage). This reflects his dominant offensive role in most of his career.

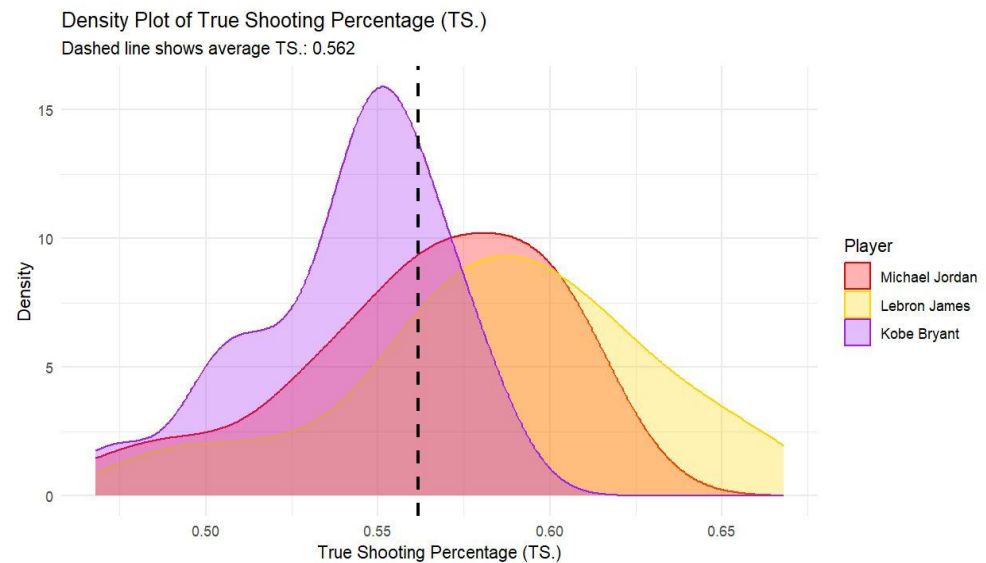
Distribution of Usage Rate (USG.) for 3 Players





TRUE SHOOTING PERCENTAGE (TS)

True shooting percentage (TS%) is an advanced shooting efficiency metric that accounts for field goals, 3-point field goals, and free throws. It provides a more complete measure of scoring efficiency than traditional field goal percentage.



Kobe Bryant (Purple Curve)

TS% distribution is centered left of the average line, peaking around 0.54–0.56. Indicates he was less efficient as a scorer than both LeBron and Jordan. This matches broader analysis that Kobe was a high-volume scorer but not elite in efficiency.

LeBron James (Yellow Curve)

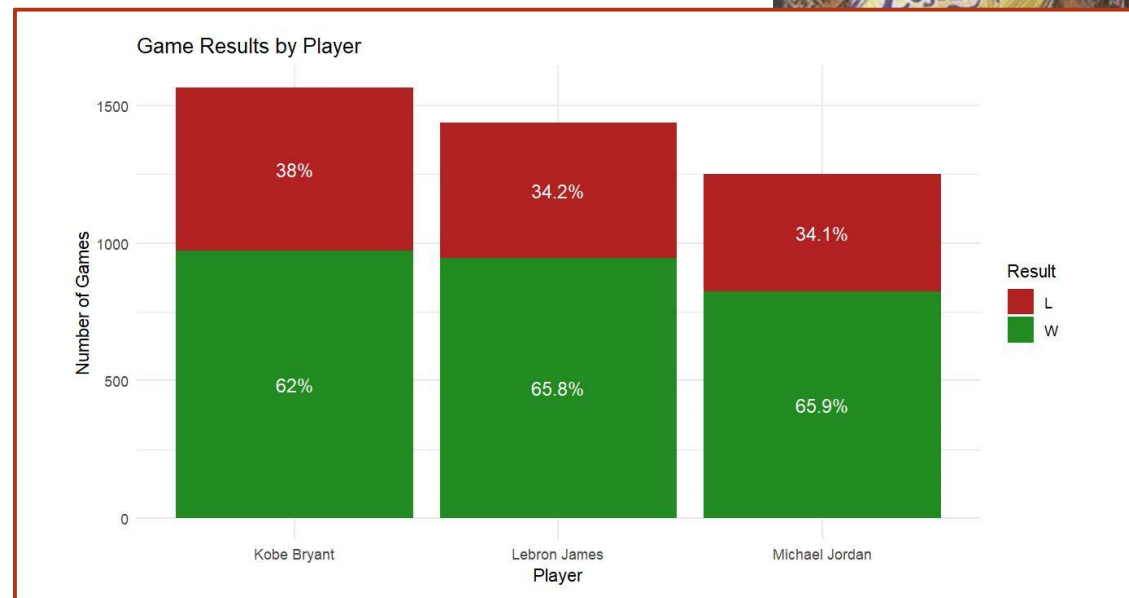
Distribution is right-shifted, peaking above the average TS%. He has the highest TS% overall of the three. Indicates very efficient scoring, likely due to shot selection, finishing ability, and free throw frequency..

Michael Jordan (Red Curve)

His peak is around the average line, slightly to the right. Suggests above-average efficiency, especially impressive given his era and volume. More balanced TS% distribution compared to LeBron's skew toward higher efficiency.

WINS & LOSSES

- Each player's bar is divided into **two sections**:
- **Green (Wins):** Percentage of games won
- **Red (Losses):** Percentage of games lost
- **Numbers inside the bars** show exact percentages



Kobe Bryant

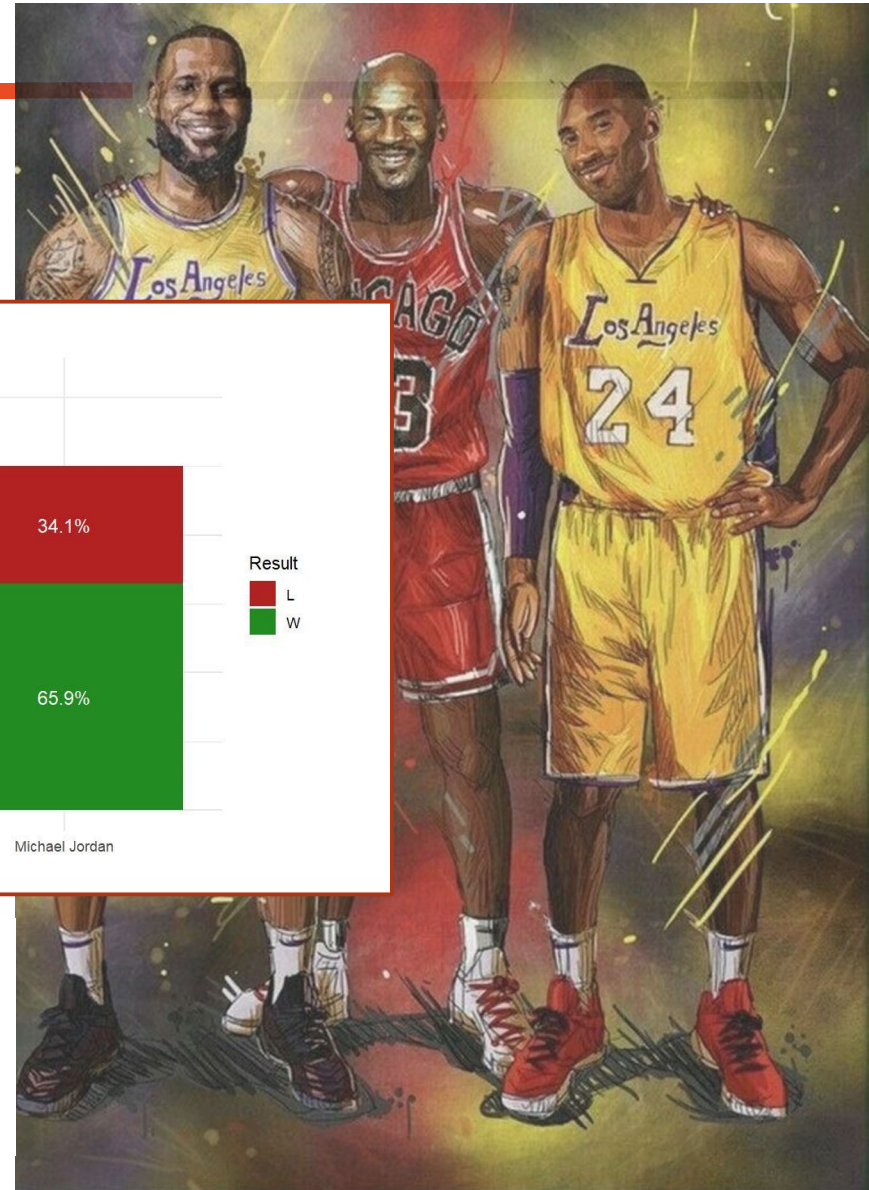
This is the lowest win percentage of the three players. Reflects a long career with some rebuilding seasons and late-career struggles (post-Shaq era, injuries).

LeBron James

Very high win percentage, slightly below Jordan. Reflects long-term consistency and success, especially during his Miami and Cleveland years.

Michael Jordan

The highest win percentage of the three, though just barely above LeBron. Jordan played fewer total games than the other two and had most of his career during the Bulls' dynasty years.



OVERALL ANALYSIS



To evaluate and compare the overall greatness of Michael Jordan, LeBron James, and Kobe Bryant, I considered both **individual performance** and **team contributions** using the metrics categorized for player impact [PTS, AST, TRB, BLK, STL, TOV, FG, PER], and team impact [WS, USG, TS, W/L].

Kobe Bryant

The Fearless Competitor

Usage Rate: Very high with a wide spread, indicating phases of carrying teams alone.

True Shooting %: Lowest among the three, suggesting volume over efficiency.

Win Percentage: Strong (62%), but slightly behind due to rebuilding years and tough late-career stretches.

5: NBA Championships

LeBron James

The Complete All-Around Force

Usage Rate: High, but more balanced, showing his role as both scorer and facilitator.

True Shooting %: Best among the three, demonstrating elite scoring efficiency.

Win Percentage: Nearly identical to Jordan's (65.8%), showing sustained success across multiple franchises.

4: NBA Championships

Michael Jordan

The Relentless Scorer Winner

Winner Usage Rate (USG%): Consistently high, reflecting his dominant offensive role.

True Shooting % (TS%): Above average, especially impressive given his volume and era.

Win Percentage: Highest at 65.9%, with sustained team success during peak years.

6: NBA Championships



G(GREATEST)
O(OF)
A(ALL)
T(TIME)



GOAT

